

The datasheet is particularly important in revealing the dependence of different parameters on operating temperature and also in revealing the dependence of one parameter on another related parameter. Forward and reverse current specifications being functions of forward and reverse voltage drops, respectively, are some examples. This helps the designer in predicting the behaviour of the device under different operating conditions.

role in the functioning of switching diodes. Smaller value of diode capacitance will give faster switching times. It is typically a few pico-farads in switching diodes.

For general-purpose diode 1N4001, the total capacitance is 15 pF for a reverse voltage of 4V, whereas in the case of 1N914, which is a switching diode, transition capacitance is less than 1 pF at the same value of reverse-bias voltage. The datasheets of switching diodes contain the characteristic curve showing variation of the diode's capacitance with the applied reverse bias (Fig. 8).

Ampere-square seconds (I^2t). It indicates the sub-cycle current capability of diodes when used as a rectifier. It is usually specified for one complete cycle of 50/60Hz operation. It can be calculated from one cycle of peak surge current rating of the diode as the product of the square of the RMS

equivalent of the surge current and the time period of the AC waveform. It is useful when choosing an appropriate diode rectifier.

The datasheets generally do not directly specify the I^2t rating; it is generally specified in the form of forward surge current for a sine wave of a specified time period. For instance, the datasheet of diode 1N4001 specifies the non-repetitive peak forward surge current of 30A for 8.3ms single half-sine-wave corresponding to 60Hz AC mains (Table II). The I^2t rating equals $(30/\sqrt{2})^2 \times 8.3 \times 10^{-3} = 3.735 \text{ A}^2\text{s}$.

I^2t rating allows designers to estimate the peak forward surge current that the device shall be able to handle for any given value of sub-cycle time period. For example, 1N4001 shall be able to handle a non-repetitive forward peak current of more than 6A for a 2ms period.

Reverse recovery time (t_r). When

TABLE II
Absolute Maximum Ratings*

Symbol	Parameter	Value	Units
$I_{(AV)}$	Average rectified current 0.375" lead length @ $T_A = 75^\circ\text{C}$	1.0	A
I_{FSM}	Non-repetitive peak forward surge current 8.3ms single half-sine-wave Superimposed on rated load (JEDEC method)	30	A
P_D	Total device dissipation Derate above 25°C	2.5 20	W mW/ $^\circ\text{C}$
$R_{\theta JA}$	Thermal resistance, junction to ambient	50	$^\circ\text{C/W}$
T_{stg}	Storage temperature range	-55 to +175	$^\circ\text{C}$
T_J	Operating junction temperature	-55 to +150	$^\circ\text{C}$

*These ratings are limiting values above which the serviceability of any semiconductor device may be impaired.

$T_A = 25^\circ\text{C}$ unless otherwise noted

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